

June 18, 2026

To the Editorial Board
Physical Review Research

Dear Editors,

Please consider the enclosed manuscript, “A two-feedback Vicsek–Couzin flock: dense-phase noise rectification as the unifying action of density-gated motility–noise coupling,” for publication in *Physical Review Research* as an original research article. The Vicsek model fixes the self-propulsion speed and the angular-noise distribution as global constants; a density-dependent speed (which drives motility-induced phase separation) and heavy-tailed Lévy reorientations have each been relaxed separately. We ask what happens when both are gated together by one density-dependent sigmoid in the local neighbour count, isolating each channel with a four-way ablation on a single code path.

Heavy-tailed noise alone grows no size-dependent susceptibility; only the motility channel builds a phase-separated dense phase, inside which the noise-shape channel rectifies its Cauchy kicks into Gaussian ones. What at first looks like a super-additive cooperation is, we show with a matched-noise control, a single effect: equalising the effective dense decorrelation collapses the susceptibility excess (Δ_9 from $+0.45$ to $+0.02$) and reverses the static $g(r)$ and dynamic $C(\tau)$ gaps, so the two channels are additive and the noise-shape feedback simply lowers the dense effective noise. The robust results are a two-regime split that reproduces across three densities, a density-gated Lévy-to-Gaussian conversion of dense-phase turning, and a decoupled-sigmoid dissection pinning the motility threshold as the sole, metric-kernel control.

The work suits *Physical Review Research* and its broad, open-access remit: it is a careful, well-controlled active-matter study whose central result is clarifying and whose null results (a passive noise-shape threshold, negligible hysteresis, no extended patch lifetime, a topological-kernel breakdown, a falsified mean-density prediction) are reported rather than buried, with the finite-size slopes framed as crossover rates, not asymptotic exponents. All code and data are openly available at <https://github.com/yayayou47/two-feedback-vicsek> for bit-for-bit reproduction of every figure. The manuscript is original, is not under consideration elsewhere, and its two authors declare no competing interests; one supplementary-material PDF accompanies it.

Thank you for considering our submission.

Sincerely,



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